

Owen Skriloff

CONTACT INFORMATION	University of Chicago Pritzker School of Molecular Engineering 177 College Ave, Medford, MA 02155	https://oskril01.github.io owenskriloff@me.com
EDUCATION	PhD, University of Chicago Pritzker School of Molecular Engineering <i>Molecular Engineering</i> Advisors: Samantha Riesenfeld & Frederic Chong MS, Tufts University School of Arts and Sciences <i>Mathematics</i> Concentration in Probability and Statistical Learning Theory BS, Tufts University School of Engineering <i>Chemical and Biological Engineering and Mathematics</i> Highest Thesis Honors, Summa Cum Laude	(In Progress) May 2026 May 2025
AWARDS	Benjamin G Brown Scholarship (<i>Research Excellence</i>) Christos Georgakis Award (<i>Best Thesis</i>) Chemical Engineering Summer Research Grant Regeneron Science Talent Search Top Scholar National Junior Science and Humanities Symposium: 2nd Place Physical Sciences	2025 2025 2024 2020 2020
PUBLICATIONS	[1] Skriloff, O. 2025. Dimensionless Controls of Plasticity Under Alternating Tasks: From Evolutionary Biology to Continual Learning. <i>Under Review</i> . [2] Skriloff, O., D. M. Stoukides, and E. S. Tzanakakis. 2025. Augmentation Models of Stem Cell Culture Data for the Application of Machine Learning. <i>Biotechnology and Bioengineering</i> 0: 1-11. https://doi.org/10.1002/bit.70094 .	
CONFERENCES	[1] O. Skriloff, D. Stoukides, E. Tzanakakis. Data Augmentation Models for Machine Learning Applications in Stem Cell Bioprocessing. <i>AIChE Annual Meeting</i> , 2025. Speaker . [2] O. Skriloff, D. Stoukides, E. Tzanakakis. Data augmentation models for stem cell bioprocessing. <i>Cell Culture Engineering XIX (ECI)</i> , 2025. Poster .	
EMPLOYMENT	Tzanakakis Research Group <i>Researcher</i> • Engineered and trained three physics-informed ML models of stem cell growth dynamics. • Built novel data pipeline with Monte Carlo simulation and feature importance to quantify and visualize representational alignment between models and known biological behavior. • Determined optimal model class and design procedure for stem cell manufacturing. Itaconix Corporation <i>Summer Associate Engineer</i> • Commissioned, purchased parts for, and assembled a 100L SS reactor with vacuum pump. • Scaled an esterification process from 250mL to the first pilot production at 50L. • Developed HPLC and GPC methods for 200+ samples and supported GPC installation. Araca Inc. <i>Laboratory Technician</i> • Conducted 25+ chemical mechanical planarization (CMP) and post-CMP cleaning experiments in a Class 100 cleanroom to optimize CMP conditions for semiconductor manufacturing. • Measured 250+ wafers, mixed slurries, programmed polishers, and tested new CMP products. • Analyzed data in MATLAB, Excel, and LabVIEW to prepare 25+ presentations visualizing experimental results to inform client process improvements.	Medford, MA <i>Jan 2024 - May 2025</i> Stratham, NH <i>Jun 2023 - Aug 2023</i> Tucson, AZ <i>Sep 2020 - Dec 2020</i>

Temple University Smart Biomaterials Lab*Researcher*

Philadelphia, PA

Mar 2019 - Aug 2020

- Developed and performed methodology to quantify a relationship between stress and mineral deposits generated by electric charges from a piezoelectric composite using simulated body fluid, incubator, and actuator.
- Authored research report: An *in-vitro* evaluation of the relationship between stress and mineralization through the use of a piezoelectric barium titanate composite.

SKILLS**Computer Languages**

Python, MATLAB, LaTeX, HTML.

Scientific Computing

Neural networks (PyTorch, TensorFlow), data science (scikit-learn, NumPy, pandas), data visualization (seaborn, matplotlib), version control (Git, Unix, Bash), object-oriented programming, Monte Carlo simulation, optimization, finite element solvers.

Engineering

Reaction design and scale-up, ChemStation, HPLC and GPC, chemical mechanical planarization, reaction-kinetic modeling, process control modeling, SuperPro Designer.

**PERSONAL
INTERESTS**

Outside of academics, I'm an avid reader and artist. I love to draw, paint, play saxophone, and spend time traveling and in museums. I also enjoy skiing, running, hiking, and doing flips.